



NCCER Module 22101



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Orientation to The Trade





Section One

Orientation to the Trade

Objective

1. Explain the role of heavy equipment operators in the construction industry.
 - a. Explain the uses of heavy equipment.
 - b. Identify and describe the different types of heavy equipment.
 - c. Describe the career opportunities associated with heavy equipment operation.
 - d. Describe the training available to heavy equipment operators.



Performance Tasks

Under the supervision of an instructor, trainees should be able to do the following:

There are no Performance Tasks associated with this section.



1.0.0 Heavy Equipment Operator Training

The purpose of Level One is to familiarize you with:

- Heavy equipment and its controls
- Maintenance activities
- Safety precautions and procedures



(A) BULLDOZER



(B) ARTICULATED OFF-ROAD DUMPTRUCK



(C) EXCAVATOR



(D) BACKHOE LOADER



(E) SKID STEER



(F) MOTOR GRADER



1.1.0 Heavy Equipment Applications

Prepare the job site for construction:

- Clear structures, plants, and boulders.
- Level the job site.
- Dig trenches for utilities.
- Prepare the site for paving and landscaping.



(G) VIBRATORY ASPHALT COMPACTOR



(H) STRAIGHT MAST ROUGH TERRAIN FORKLIFT



(I) TANDEM POWERED SCRAPER



(J) UTILITY TRACTOR



(K) OFF-HIGHWAY HEAVY CONSTRUCTION TRUCK



1.2.0 Types of Heavy Equipment (1 of 4)



- Clearing and grubbing is done with a dozer.
- The site is brought to subgrade or rough grade by moving soil.
- Material may be trucked in or out to achieve the required grades.

1.2.0 Types of Heavy Equipment (2 of 4)

Excavators and backhoes dig foundations and trenches.



1.2.0 Types of Heavy Equipment (3 of 4)

Scrapers and soil compactors prepare the soil and gravel for paving.



(A)



(B)



(C)

1.2.0 Types of Heavy Equipment (4 of 4)

- Paving equipment is brought in to finish the road.
- A heavy asphalt compactor will be used to pack and smooth asphalt.



1.3.0 Career Opportunities in Heavy Equipment

- Building construction
- Roads
- Bridges
- Tunnels
- Mining projects
- Power-generating facilities, including wind farms, solar farms, and biomass facilities



1.3.1 Getting Started

- Operators start out learning smaller equipment such as forklifts.
- Many companies start trainees on interactive computer simulations.
- After successful computer training, the trainee will move on to hands-on training.
- Advanced operators can later move on to heavier equipment such as dozers, excavators, and graders.



1.3.2 Other Job-Related Opportunities

- Instructors in a company's own training program
- Equipment operator instructors at an independent school
- Sales or management roles at a heavy equipment rental company



1.3.3 Supervisory and Management Positions

- Crew leader
- Foreman
- Equipment coordinator
- Equipment superintendent
- Project manager
- Project superintendent



1.3.4 Self-Employed Contractor

- Many construction companies hire independent operators, either for the entire job or during peak times.
- The way to be successful as an independent operator is to develop a reputation for being reliable and doing quality work.



1.4.0 Standardized Training By NCCER

NCCER offers:

- Standardized education materials
- A tracking system for training progress
- Accreditation
- Instructor certification
- Skills assessments



1.4.1 Apprenticeship Training

All approved apprenticeship programs have on-the-job (OJL) learning and classroom instruction.

New U.S. Department of Labor regulations give three different options for apprenticeship programs:

- Traditional, time-based programs
- Competency-based programs
- Hybrid programs



1.4.2 Youth Training and Apprenticeship Programs

- Students may begin training while in high school.
- Students may be able to enter the industry with higher pay and experience.
- Students may transfer some training credits to colleges and universities.



Wrap Up - Trade Terms

Demolition: The destruction and removal of a structure.

Grading: Establishing the proper level or slope of the ground.

Infrastructure: A system of public works such as highways, bridges, tunnels, and dams.

On-the-job learning (OJL): Job-related learning acquired while working.

Policies: General statements establishing guidelines for a specific activity.



1.0.0 Section Review Question 1

1. Which of these machines is used primarily in the final stage of site preparation?
 - a. Motor grader
 - b. Excavator
 - c. Backhoe-loader
 - d. Dozer



1.0.0 Section Review Question 1 Answer

1. Which of these machines is used primarily in the final stage of site preparation?
 - a. Motor grader**
 - b. Excavator
 - c. Backhoe-loader
 - d. Dozer



1.0.0 Section Review Question 2

2. Which of these machines is likely to be used for trenching?
- a. Dozer
 - b. Compactor
 - c. Excavator
 - d. Scraper



1.0.0 Section Review Question 2 Answer

2. Which of these machines is likely to be used for trenching?
- a. Dozer
 - b. Compactor
 - c. Excavator**
 - d. Scraper



1.0.0 Section Review Question 3

3. About how many heavy equipment operators are there in the US?
- a. Less than 100,000
 - b. Roughly 200,000
 - c. More than 400,000
 - d. More than 1 million



1.0.0 Section Review Question 3 Answer

3. About how many heavy equipment operators are there in the US?
- a. Less than 100,000
 - b. Roughly 200,000
 - c. More than 400,000**
 - d. More than 1 million



1.0.0 Section Review Question 4

4. In addition to on-the-job learning, what is the minimum number of annual classroom hours a registered apprentice must complete in a traditional apprenticeship program?
- a. 50
 - b. 84
 - c. 100
 - d. 144



1.0.0 Section Review Question 4 Answer

4. In addition to on-the-job learning, what is the minimum number of annual classroom hours a registered apprentice must complete in a traditional apprenticeship program?
- a. 50
 - b. 84
 - c. 100
 - d. 144**



Next...

**Section 2.0.0
Responsibilities and
Characteristics of Operators**



*Read Sections 2.0.0 through 2.3.3 and
complete the Section Review.*